

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-aa4e72ec14e852306.jsonl`  
- SHA-256: `12d59e2cb85db00e022afa0c0d043995081ffc282941fb809b3144793b07eae4`  
- Source modified: `2026-06-16T03:20:14+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:04.328Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/pdf/2512.01298>

****Title:**** TBT-Former: Learning Temporal Boundary Distributions for Action Localization (arXiv 2025)

****Found via:**** Single-stage TAL actionness + boundary heads, small data search

Task

1. Use WebFetch to retrieve the page content.
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Structured output only.

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I'll fetch the source and extract relevant claims.

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TBT-Former: Key Claims Extraction

(1) What is TBT-Former and Problem Statement

****Title & Authors:**** "TBT-Former: Learning Temporal Boundary Distributions for Action

Localization" by Thisara Rathnayaka and Uthayasanker Thayasivam

****Problem Addressed:**** The model tackles the challenge of precise temporal action localization, specifically focusing on learning distributions of temporal boundaries rather than point estimates, which helps address over-segmentation issues common in action detection.

(2) Methodology

The approach employs a Transformer-based architecture that learns probability distributions over temporal boundaries. Rather than predicting single boundary locations, the method models boundary uncertainty through distributional outputs, incorporating actionness scoring and boundary prediction heads to refine temporal action segment identification.

(3) Experimental Results

The document references evaluation on standard action localization benchmarks. The paper includes comparative results with citations to prior work (references b2, b4, b6 appear frequently in results sections), though specific mAP@tIoU and F1 scores are embedded within the compressed PDF content streams and not readily extractable in this format.

(4) Small/Limited Data Settings

The document structure suggests comprehensive experimental sections (pages 4-8 contain results), but specific treatment of limited-data scenarios cannot be confirmed from the readable metadata.

(5) Publication Details

- ****Venue:**** arXiv (preprint)
- ****Date:**** December 2, 2025
- ****Identifier:**** arXiv.2512.01298v1

[Binary content (application/pdf, 698.3KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579953015-t4vlj4.pdf]

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pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. user (2026-06-16T03:19:35.455Z)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
===== PAGE 1 =====

TBT-Former: Learning Temporal Boundary
Distributions for Action Localization
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Abstract?TemporalActionLocalization(TAL)remainsafun- Transformer-
basedarchitecture.Bytreatingeverymomentina
damental challenge in video understanding, aiming to identify
videoasapotentialactioncenterandusinglocalself-attention
the start time, end time, and category of all action instances
within a multi-scale encoder, ActionFormer achieved state-
within untrimmed videos. While recent single-stage, anchor-free

of-the-art results without relying on pre-defined anchors or models like ActionFormer have set a high standard by leveraging Transformers for temporal reasoning, they often struggle with explicit proposal generation stages [2]. Despite its success, two persistent issues: the precise localization of actions with the ActionFormer framework and similar models exhibit two ambiguous or "fuzzy" temporal boundaries and the effective primary limitations that motivate our work.

Fusion of multi-scale contextual information. In this paper, we first, the problem of imprecise boundary regression introduces the Temporal Boundary Transformer (TBT-Former), persists. Standard regression heads, typically optimized with an architecture that directly addresses these limitations. TBT-Former enhances the strong ActionFormer baseline with three core contributions: (1) a higher-capacity scaled Transformer Intersection over Union) loss, predict a single, deterministic backbone with an increased number of attention heads and an offset for an action's start and end times. This formulation expanded Multi-Layer Perceptron (MLP) dimension for more is inherently inflexible and struggles when action boundaries powerful temporal feature extraction; (2) a cross-scale feature are ambiguous? for instance, when an action begins or ends pyramid network (FPN) that integrates a top-down pathway with gradually. In such real-world scenarios, forcing the model to lateral connections, enabling richer fusion of high-level semantics and low-level temporal details; and (3) a novel boundary distribution. predict a single correct timestamp can lead to noisy training distribution regression head. Inspired by the principles of Generalized signals and inaccurate localization [4].

Focal Loss (GFL), this new head recasts the challenging task of Second, the mechanism for multi-scale feature fusion boundary regression as a more flexible probability distribution can be improved. The feature pyramid in ActionFormer is learning problem, allowing the model to explicitly represent constructed in a purely feed-forward manner, where strided and reason about boundary uncertainty. Within the paradigm of Transformer-based architectures, TBT-Former advances the operations generate progressively coarser feature maps [2]. formidable benchmark set by its predecessors, establishing a new This architecture lacks an explicit mechanism for coarser, level of performance on the highly competitive THUMOS14 and semantically rich features to inform and refine the finer-EPIC-Kitchens 100 datasets, while remaining competitive on the grained features at higher temporal resolutions. This limits large-scale ActivityNet-1.3. Our code is available here. the model's ability to integrate global context with precise Index Terms? Temporal Action Localization, Transformers, Video Understanding, Boundary Regression, Feature Pyramid local details, which is crucial for localizing actions of vastly Network different durations [2].

To address these challenges, we propose the Temporal I. INTRODUCTION Boundary Transformer (TBT-Former), a new model that in- The ability to automatically understand human activities introduces three targeted architectural advancements upon the from video is a cornerstone of modern computer vision, ActionFormer baseline. with applications ranging from video summarization and 1) A Scaled Transformer Backbone: We increase the surveillance to human-computer interaction [1]. A critical model's representational capacity by scaling up the task in this domain is Temporal Action Localization (TAL), Transformer encoder, employing 16 attention heads and which requires not only classifying actions but also precisely expanding the MLP hidden dimension by a factor of identifying their temporal start and end points within long, 6x. This allows the model to

capture more intricate and untrimmed videos. The evolution of TAL methodologies has long-range temporal dependencies. seen a significant shift from complex two-stage, proposal- 2) A Cross-Scale Feature Pyramid (CS-FPN): Inspired based frameworks to more streamlined and efficient single- by refined Feature Pyramid Network (FPN) structures stage, anchor-free approaches [2], [3]. [5], we incorporate a top-down pathway with lateral A seminal work in this new paradigm is ActionFormer [2], connections. This architecture facilitates a bidirectional which demonstrated the remarkable efficacy of a minimalist, flow of information, enriching fine-grained temporal

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===== PAGE 2 =====

features with high-level semantic context from coarser spatially precise information in fine-grained maps, a principle scales. we incorporate into our Cross-Scale FPN.

3) A novel Boundary Distribution Regression (BDR)

C. Boundary Modeling in Detection

Head: This is our main contribution. Drawing inspiration from the core philosophy of Generalized Focal Loss A persistent challenge in localization tasks is the inherent (GFL) [6], we reframe boundary prediction. Instead of ambiguity of object or action boundaries. Standard regression regressing a single offset value, our BDR head learns a losses, such as L1 or Smooth L1, assume a single, deterministic flexible probability distribution over a range of possible istic ground-truth location, which is an oversimplification for temporal locations. This allows the model to explicitly many real-world cases. This issue was elegantly addressed in represent boundary uncertainty, leading to more robust the object detection domain by Generalized Focal Loss (GFL) and accurate localization, especially for actions with [6]. The core insight of GFL is to abandon the assumption ambiguous transitions. of a Dirac delta distribution for a bounding box coordinate. Instead, it represents the coordinate as a flexible, arbitrary Through extensive experimentation, TBT-Former demonstrates superior performance, setting a new state-of-the-art on The final coordinate is then computed as the expectation of the THUMOS14 [7] and EPIC-Kitchens 100 [8] datasets and this distribution, and a continuous version of Focal Loss is achieving highly competitive results on ActivityNet-1.3 [9]. used for optimization [6]. This allows the model to represent Our work shows that explicitly modeling boundary uncertainty its uncertainty; a sharp distribution indicates high confidence, is a powerful and promising direction for advancing temporal while a flatter one signals ambiguity. In the TAL domain, action localization. TriDet [4] also explored modeling boundaries as distributions.

II. RELATED WORK Our proposed Boundary Distribution Regression (BDR) head is conceived as a direct and principled adaptation of the

A. Single-Stage Temporal Action Localization

GFL philosophy to the temporal domain, providing a novel Early TAL method often adopted a two-stage pipeline: first and effective solution for handling the pervasive issue of generating a sparse set of candidate temporal proposals, and ambiguous action boundaries in videos. then classifying and refining them [10], [11]. Models like BMN [10] achieved strong performance in proposal generation III. OVERVIEW OF ACTION FORMER but were often computationally intensive and not end-to- Our work builds upon the strong foundation of Action-

end trainable. The field has since moved towards single-stage, effective single-stage, stage detectors, which perform localization and classification anchor-free framework for temporal action localization. The simultaneously in a single pass. A key innovation within this core philosophy of ActionFormer is to treat every moment paradigm is the anchor-free approach, which discards pre-defined anchor windows and instead treats every temporal the complex task of structured output prediction into a more location on a feature map as a potential action candidate [2]. manageable sequence labeling problem. An overview of the This design philosophy, exemplified by models like A2Net, ActionFormer architecture is provided in Figure 1.

GTAN, and notably ActionFormer, simplifies the architecture The model follows a standard encoder-decoder structure. and has led to significant gains in both efficiency and accuracy The encoder is a multi-scale Transformer network that pro-

[2]. TBT-Former builds directly upon this single-stage, anchor-free foundation, inheriting its efficiency while targeting its score self-attention to model temporal dependencies efficiently, even representational limitations. for long videos. A key design choice is the creation of a feature pyramid, where strided operations within the Transformer

B. Multi-Scale Representations in Video

blocks generate feature maps at progressively coarser temporal

Actions in videos can span from a few seconds to several minutes. This multi-scale representation allows the model minutes, making multi-scale feature representation a critical to effectively detect actions of varying durations [2].

component of any robust TAL model. The concept of a Feature Pyramid Network (FPN), originally developed for object works that act as prediction heads. These heads are shared

detection, has been widely adopted to handle this variation across all levels of the feature pyramid. For each temporal

[5]. An FPN creates a hierarchy of feature maps at different location on the pyramid, two parallel branches perform the

temporal resolutions, allowing the model to detect both short main tasks: a classification head predicts the probability for

and long action effectively. The baseline ActionFormer model

each action class, and a regression head estimates the temporal

implements a simple, feed-forward pyramid where deeper distances from the current moment to the action's start and

layers have coarser temporal resolution [2]. However, more end boundaries. The model is trained end-to-end using a

advanced structures, such as the one explored in [5], employ combination of Focal Loss [12] for classification and a DIOU

top-down pathways and lateral connections to explicitly merge

loss [13] for boundary regression [2]. This is simple yet powerful

features across scales. This allows high-level semantic information from coarse feature maps to flow back and enrich the baseline for further innovation in the field.

==== PAGE 3 =====

Fig. 1: An overview of the baseline ActionFormer architecture, illustrating the multi-scale Transformer encoder and

Fig. 2: Conceptual illustration of the TBT-Former architecture, the lightweight convolutional decoder with classification and highlighting the Scaled Backbone, Cross-Scale FPN (CS-

regression heads [2]. FPN), and Boundary Distribution Regression (BDR) Head

modules as key enhancements to the ActionFormer baseline.

IV. METHOD

A. Architecture Overview 4x expansion. This substantial increase in parameters provides the model with greater representational power, enabling it to

TBT-Former builds upon the robust and efficient anchor-capture more nuanced patterns in the temporal dynamics of free architecture of ActionFormer [2]. The baseline model actions, which is particularly beneficial for complex scenes consists of a Transformer-based encoder that processes a and diverse datasets.

sequence of video features, followed by a lightweight convolutional decoder with separate classification and regression

C. Cross-Scale Feature Pyramid (CS-FPN)

heads. Our proposed TBT-Former retains this overall structure

The original ActionFormer builds its feature pyramid

but introduces three significant enhancements: a scaled-up

$\{P_1, P_2, P_3, P_4\}$ in a simple feed-forward manner, where each

backbone for increased capacity, a cross-scale feature pyramid

level is generated by downsampling the previous one [2]. This

mid for improved multi-scale fusion, and a novel regression

design prevents the flow of information from semantically rich,

head for probabilistic boundary modeling. Figure 2 illustrates

coarse-grained levels back to temporally precise, fine-grained

the overall architecture of TBT-Former, highlighting our key

levels. To remedy this, we introduce a Cross-Scale Feature

modifications.

Pyramid (CS-FPN), inspired by the design in [5]. The overall

B. Scaled Transformer Backbone architecture and the specific fusion operation are illustrated in

To enhance the model's ability to learn complex temporal Figure 4.

relationships and long-range dependencies, we first scale the Our CS-FPN enhances the feature

hierarchy with a top-

capacity of the Transformer encoder. While the baseline Ac- down pathway and lateral

connections. Let $\{C_2, C_3, C_4, C_5\}$

tionFormer uses a standard configuration for its Transformer be the output feature maps from

different stages of our scaled

blocks, we modify the key hyperparameters to create a more backbone, with decreasing temporal

resolution. The pyramid

powerful backbone, as illustrated in Figure 3. Specifically, construction proceeds as follows:

within each Multi-Head Self-Attention (MSA) block, we in- 1) The coarsest feature map, C_5 , is

first passed through a

5
create the number of parallel attention heads from the base- 1×1 convolutional layer to produce the initial pyramid

line's 8 to 16. Furthermore, in the subsequent feed-forward map, P_5 .

5
network (FFN), we expand the hidden dimension by a factor 2)

For each subsequent level i (from 4 down to 2), the cor-

of 6 relative to the input dimension, compared to the standard responding pyramid map P_i

is computed by merging the

i
===== PAGE 4 =====

Fig. 4: The architecture of the Cross-Scale Feature Pyramid

Network (CS-FPN). The fusion operation merges features

from the bottom-up backbone pathway with the semantically

richer features from the top-down pathway via lateral connec-

tions and element-wise addition.

Motivation: A standard regression head predicts a single

float value for the start offset, ds , and end offset, de . This

formulation implicitly assumes that there is one precise,

unambiguous boundary location. However, in reality, action

boundaries are often fuzzy. Forcing the model to regress to

a single point in such cases provides a noisy and unstable

learning signal [4].

Fig. 3: The architecture of our Scaled Transformer Backbone Probabilistic Formulation: Inspired

by Generalized Focal

unit. Compared to the baseline ActionFormer, we introduce Loss (GFL) [6], we reformulate this

deterministic regression

two key enhancements for increased representational capacity: problem as a probabilistic one. Instead

of predicting a sin-

(1) the number of heads in the Multi-Head Attention module λ , our BDR head predicts a discrete probability

7. assistant (2026-06-16T03:19:37.964Z)

I have the core content. Let me get the experimental results pages (5-8) for the reported numbers.

8. user (2026-06-16T03:19:43.360Z)

TOTAL PAGES: 8

===== PAGE 5 =====

the standard Focal Loss (L) for classification and our DFL OnEPIC-Kitchens100, presented in Table III, TBT-Former

cls
for the start and end boundaries: consistently outperforms ActionFormer and prior methods for both Verb and Noun classification, achieving average

$L = L_{cls} + \lambda(L_{verb} + L_{noun})$ (4)

cls DFL s gt DFL e gt mAP improvements of 1.0% and 1.2% over ActionFormer, where λ is a balancing hyperparameter. This loss structure respectively [2]. The substantial gain, especially for Noun

provides a more stable and accurate learning objective for the classification, suggests that our BDR head is particularly

challenging task of temporal boundary regression. effective for localizing the short, object-centric actions that

are prevalent in egocentric video.

V. EXPERIMENTS

C. Ablation Studies

A. Datasets and Evaluation Metrics

To validate our design choices and quantify the contribu-

We evaluate TBT-Former on three widely-used and challenging TAL benchmarks: extensive set of ablation studies on the THUMOS14 dataset.

1) THUMOS14 [7]: This dataset is a standard benchmark

Summary of Component Contributions: We first present

for TAL, containing 20 action categories in over 400 a summary of how each of our proposed components con-

untrimmed videos. It is characterized by long videos with tributes to the final performance. As shown in Table IV,

a high density of action instances, making it particularly each modification provides a clear and positive impact. The

well-suited for evaluating localization precision. Scaled Backbone, CS-FPN, and BDR Head contribute +0.4,

2) ActivityNet-1.3 [9]: A large-scale dataset featuring 200

+0.3, and +0.8 mAP, respectively. When combined, they work

action classes in approximately 20,000 videos. Its scale synergistically to achieve a total improvement of +1.2 mAP

and diversity make it an excellent testbed for model over the strong ActionFormer baseline. generalization and robustness. 2) Effect of Local Attention Window Size: The original

3) EPIC-Kitchens 100 [8]: The largest dataset for ego- ActionFormer paper used a local attention window of size

centric action understanding. It contains numerous short, 19 for THUMOS14 [2]. Given that our Scaled Transformer

fine-grained actions defined by verb-noun pairs. Its Backbone has a higher representational capacity, we hypoth-

unique characteristics test a model's ability to perform esized that it could effectively leverage a larger temporal

precise localization on brief, object-centric interactions.

context. We experimented with several window sizes, with the

For all datasets, we follow standard evaluation protocols and results shown in Table V.

Performance steadily improves as

report the mean Average Precision (mAP) at various temporal the window size increases from 19 to 30, where it peaks at

Intersection over Union (tIoU) thresholds [2], [7]. an average mAP of 68.0%. Beyond this,

performance begins to plateau, suggesting that a window size of 30 provides the optimal balance between contextual reach and model focus for

B. Main Results

We compare TBT-Former with the ActionFormer baseline our enhanced architecture. and other state-of-the-art methods. 3) Design of the Feature Pyramid: We validate the feature Performance on THUMOS14: As shown in Table I, TBT- pyramid design inherited from ActionFormer [2]. As shown Former establishes a new state-of-the-art on the THUMOS14 in Figure 5, performance is highly sensitive to the number of benchmark. It achieves an average mAP of 68.0%, surpassing pyramid levels. A single-level model performs poorly (47.6% the strong ActionFormer baseline by 1.2 absolute points [2]. avg. mAP). Performance consistently increases with more The inclusion of prior single-stage (GTAN, A2Net) and two- levels, saturating at 6 levels, which confirms that a deep, stage (BMN) methods further highlights the significant and multi- scale pyramid is critical for capturing actions of varying consistent progress made, with TBT-Former outperforming all durations. competitors across all tIoU thresholds [2]. This demonstrates 4) Analysis of Loss Weight ? : We analyze the balance reg the effectiveness of our architectural enhancements, particu- between the classification loss and our new DFL regression larly on a dataset that heavily rewards precise localization. loss, controlled by the hyperparameter? . As shown in Table reg Performance on ActivityNet-1.3 and EPIC-Kitchens100: VI, the model is robust to a range of values, but performance Tables II and III summarize our results on the ActivityNet-1.3 peaks at ? = 1.0. This confirms that an equal weighting reg and EPIC-Kitchens 100 validation sets, respectively. between the classification and boundary distribution regression On the large-scale ActivityNet-1.3 dataset, TBT-Former objectives provides the optimal trade-off for our model. achieves highly competitive performance. As shown in Table 5) Alternative Backbone Architectures: Finally, we ex- II, our model is on par with the strong ActionFormer baseline, plored replacing the Transformer-based backbone of the base- achieving an average mAP of 36.8%. While there is a minor line model with two alternative architectures to understand dip at the tIoU=0.5 threshold, our model shows improved per- their suitability for TAL. The results are summarized in Table formance at the stricter threshold of tIoU=0.75 and tIoU=0.95 VII. [2]. This indicates superior localization precision for well- Hybrid Transformer-Mamba Backbone: We designed defined actions and demonstrate that our architectural changes a hybrid encoder that uses Transformer blocks for high- maintain strong performance on large-scale benchmarks. resolution feature maps and Mamba blocks for coarser, low- ===== PAGE 6 =====

TABLE I: Comparison with state-of-the-art methods on the THUMOS14 test set. TBT-Former achieves the highest performance across all metrics.

Model Type	Avg.mAP	mAP@0.3	mAP@0.4	mAP@0.5	mAP@0.6	mAP@0.7
GTAN[2] Single-Stage	57.8	70.4	63.3	57.8	47.2	33.5
A2Net[2] Single-Stage	58.6	70.6	65.0	59.9	51.3	37.5
BMN[2] Two-Stage	56.0	67.5	60.9	56.0	47.4	34.9
ActionFormer[2] Single-Stage	66.8	82.1	77.8	71.0	59.4	43.9
TBT-Former(Ours) Single-Stage	68.0	82.5	79.0	72.4	60.6	45.3

TABLE II: Performance on ActivityNet-1.3 validation set. The TABLE IV: Summary of component-wise contributions on average mAP is calculated over [0.5:0.05:0.95]. *Results for THUMOS14. ActionFormer are from the original paper [2].

ModelConfiguration Avg.mAP ?

Model mAP@0.5 mAP@0.75 mAP@0.95 Avg.mAP 1 Baseline(ActionFormer) 66.8 -
PGCN[3] 48.3 33.2 3.3 31.1 2 +ScaledBackbone 67.2 +0.4
BMN[10] 50.1 34.8 8.3 33.9 3 +Cross-ScaleFPN(CS-FPN) 67.1 +0.3
G-TAD[11] 50.4 34.6 9.0 34.1 4 +BoundaryDistributionHead(BDR) 67.6 +0.8
AFSD[15] 52.4 35.3 6.5 34.4 5 FullModel(TBT-Former) 68.0 +1.2
ActionFormer[2] 54.7 37.8 8.4 36.6*
TBT-Former(Ours) 53.9 38.2 8.5 36.8

TABLE V: Ablation on the local attention window size. All experiments use the full TBT-Former model.

TABLE III: Performance on EPIC-Kitchens 100 validation

WindowSize mAP@0.5 mAP@0.7 Avg.mAP

set for Verb and Noun tasks. The average mAP is over
19(Baselinesize) 71.5 44.2 67.1
[0.1:0.1:0.5].

25 72.0 44.8 67.6

30 72.4 45.3 68.0

Task Model mAP@0.1 mAP@0.2 mAP@0.3 mAP@0.4 mAP@0.5 Avg.mAP 37 72.2 45.1 67.8

Verb BMN[10] 10.8 9.8 8.4 7.1 5.6 8.4

G-TAD[11] 12.1 11.0 9.4 8.1 6.5 9.4

ActionFormer[2] 26.6 25.4 24.2 22.3 19.1 23.5

TBT-Former(Ours) 27.2 26.7 25.6 22.9 20.1 24.5

Noun BMN[10] 10.3 8.3 6.2 4.5 3.4 6.5 y(t)viaalattentstateh(t).Itsdiscretizedformisarecurrence:

G-TAD[11] 11.0 10.0 8.6 7.0 5.4 8.4

ActionFormer[2] 25.2 24.1 22.7 20.5 17.0 21.9

TBT-Former(Ours) 26.4 25.9 23.8 21.7 17.9 23.1

$h = A^-h + B^-x$

$k \geq 1$ k

(5)

$y = Ch$

$k \leq k$

resolutionmaps.MambaisarecentStateSpaceModel(SSM) Mamba?sinnovationismakingtheparameters B^- and C^-
input-

architecturewithlinear-timecomplexity,makingitefficientfor
dependent, allowing it to selectively focus on parts of the
longsequences[14].ThecoreofMambaistheSelectiveState
sequence [14]. Our hybrid architecture, shown in Figure 6(a),
Space Model (SSSM), which maps an input $x(t)$ to an output
achieved a competitive 65.8% avg. mAP.

SGP-based Backbone: We also implemented a fully-
convolutionalbackboneusingtheScalable-GranularityPercep-
tion (SGP) layer from [4]. The SGP layer, shown in Figure
6(b), replaces self-attention with two parallel branches: an
instant-level branch to enhance feature discriminability and
a window-level branch with parallel convolutions to capture
multi-granularity context. The SGP operation is formulated as
[4]:

$f = \gamma(x)FC(x) + \gamma(x)(\text{Conv}(x) + \text{Conv}(x)) + x$

SGP w kw

(6)

where $\gamma(x)$ and $\gamma(x)$ are gating functions, and FC denotes a
fully-connected layer. This model achieved 66.2% avg. mAP.
While both alternatives show strong performance, they do
not surpass the optimized Transformer baseline. This sug-
Fig. 5: Ablation on the number of feature pyramid levels.
gests that the local self-attention mechanism remains highly
Performance across different metrics (mAP@0.5, mAP@0.7,
effective for temporal feature extraction in TAL, though the
andAvg.mAP)peaksat6levels,indicatingthisistheoptimal
competitive results of SGP and Mamba highlight them as
depth for the pyramid in our architecture.
promising directions for future research.

==== PAGE 7 ====

(a) Hybrid Conv-Transformer-Mamba encoder. (b) SGP-based encoder.

Fig. 6: Architectures of the alternative backbones explored in our ablation study. (a) The

hybrid approach begins with convolutional embedding and an SE block, followed by a stem of Transformer blocks. The branch creates a feature pyramid using Transformer blocks at fine-grained levels and Mamba blocks at coarse-grained levels. (b) The SGP-based encoder is built from SGP blocks, with a two-branch design for feature discrimination and multi-granularity context.

TABLE VI: Ablation on the regression loss weight ? .

reg ActionFormer established a formidable performance benchmark. Our results demonstrate that TBT-Former successfully

?reg mAP@0.5 mAP@0.7 Avg.mAP

advances this benchmark, setting a new level of performance

0.2 70.1 39.8 65.0

on the highly competitive THUMOS14 and EPIC-Kitchens

0.5 71.4 41.7 66.4

1.0 71.0 43.6 66.9 100 datasets, while remaining competitive on the large-scale

2.0 69.7 43.1 66.3 ActivityNet-1.3. Our most significant finding is that explicitly

5.0 68.8 42.5 65.1

modeling temporal boundaries as probability distributions is a powerful and promising paradigm.

TABLE VII: Ablation on alternative backbone architectures,

Furthermore, our ablation studies on alternative backbones, compared to the ActionFormer baseline.

such as the convolutional SGP and the SSM-based Mamba

architectures, revealed their potential as efficient and compet-

BackboneType mAP@0.5 mAP@0.7 Avg.mAP

itive alternatives to self-attention. While they did not surpass

Transformer(Baseline) 71.0 43.9 66.8

our optimized Transformer-based model in this work, their

HybridAttn-Mamba 69.5 42.9 65.8

SGP(Convolutional) 70.1 43.2 66.2 strong performance indicates a valuable direction for future

research. Subsequent work could focus on refining these

architectures or developing more sophisticated hybrid models,

VI. CONCLUSION potentially unlocking further gains in both performance and computational efficiency.

In this paper, we introduced TBT-Former, a novel single-

stage, anchor-free model for temporal action localization. Our REFERENCES

work makes three key contributions: a higher-capacity scaled

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9. assistant (2026-06-16T03:19:51.487Z)

I now have the full paper content including all results tables, ablations, and references. This is a primary research paper (arXiv preprint, University of Moratuwa) directly relevant to dimension 1 (windowing/boundaries) of the research question.

Key relevant facts:

- TBT-Former builds on ActionFormer with a Boundary Distribution Regression (BDR) head that models boundaries as probability distributions (GFL-inspired) to handle fuzzy/ambiguous boundaries directly relevant to the over-merge/segmentation problem.
- Reported gains are modest absolute (+1.2 avg mAP on THUMOS14: 66.8 → 68.0; BDR head alone +0.8 mAP). EPIC-Kitchens +1.0/+1.2 mAP. ActivityNet competitive/slight dip at tIoU=0.5.
- It is NOT a small-data method; it scales up capacity (16 heads, 6x MLP expansion) and is trained on large benchmarks (THUMOS14 ~400 videos, ActivityNet ~20k, EPIC-Kitchens). This is a caveat for the ~6-16 video constraint.
- It cites TriDet for boundary distribution modeling in TAL, and explores Mamba/SGP backbones (future directions).

Now I'll return the structured output.

10. user (2026-06-16T03:20:07.216Z)

Structured output provided successfully

11. assistant (2026-06-16T03:20:14.175Z)

Extracted and verified claims from TBT-Former (arXiv 2512.01298), a primary research preprint by Rathnayaka & Thayasivam (University of Moratuwa), published December 2025. The structured output has been returned successfully.

Summary of relevance to the research question (Dimension 1 ? windowing & boundaries): The paper's central contribution, a GFL-inspired Boundary Distribution Regression head that models temporal boundaries as probability distributions over discrete bins, is the most directly applicable idea ? it targets exactly the "fuzzy boundary" problem underlying over-merged segments. However, two important caveats were flagged: (1) the payoff is modest (+0.8 mAP from the BDR head alone, +1.2 total over ActionFormer on THUMOS14), and (2) it is NOT a small-data method ? it scales up capacity (16 heads, 6x MLP) and is only evaluated on large benchmarks (THUMOS14, ActivityNet ~20k videos, EPIC-Kitchens 100), with no few-label adaptation, making direct adoption a research bet rather than a low-effort win for the ~6-16 video setting. The paper also confirms TriDet as the prior art that introduced boundary-as-distribution modeling in TAL.